MONEY AND BANKING

RESERVE MONEY / HIGH POWERED MONEY (MONETARY BASE)

RM also called the government money is produced by the Central Bank (RBI, in India) and held by the people and the banks. It is composed of:

- Currency with the public in circulation.
- Cash reserves of commercial banks: (i) cash reserves with the banks themselves and (ii) Bankers' deposits with RBI (which commercial banks maintain with RBI as reserve and consider as their '*vault*' cash).
- 'Other' deposits with RBI.

A commercial bank's required amount of reserves is equal to the required reserve ratio times the total deposits in the bank. The difference between a bank's *actual reserves* and its *required reserves* is its *excess reserves*.

Excess Reserves = [Actual Reserves – Required Reserves]

Commercial banks give loans up to the point where they can no longer do so because of the reserve requirement restriction, it means that ***banks give loans up to the point where their excess reserves are zero***.

Changes in reserve requirements affect the amount of reserves that commercial banks must keep as deposits with the RBI and consequently the amount available for lending or investing.

By *raising the reserve ratio* to be maintained by every bank, the RBI can *reduce the volume of credit* and by *lowering the reserve ratio,* it can *expand the volume of credit*.

Excess cash reserves will induce commercial banks to expand credit and reduction of cash reserves will result in contraction of credit.

CRR Reduction Schedule

- **Effective Sept. 6, 2025:** 3.75%
- **Effective Oct. 4, 2025:** 3.50%
- **Effective Nov. 1, 2025:** 3.25 %
- **Effective Nov. 29, 2025:** 3.00%

SLR: 18% [Oct. 2025]

THE MECHANISM OF MONETARY EXPANSION

The RBI requires that commercial banks retain a certain percentage of their liabilities as reserves, mainly as deposits in the Central Bank (FED/RBI).

Fed buys treasury bonds worth of $100 and consequently issues a chq drawn on itself to the seller. The seller then deposits that chq in his account in Bank A, creating $100 in liabilities for the bank and also $100 in assets for the bank, the claim on the Fed. If there is a 20% reserve requirement, Bank A can loan out $80 and must retain $20 as reserves.

The borrower spends this amount, transferring $80 to the seller's bank (**Bank B**), which in turn loans out $64. This amount is then transferred to **Bank C** and the process continues. As a result, the total increase in the money supply from the $100 reserve increase is given by

$$\Delta M = \$100 + \$80 + \$64 + \ldots$$

$$\Delta M = \$100[1 + 0.8 + (0.8)^2 + \ldots] = \frac{100}{1-0.8} = 500.$$

Balance-Sheet Effects of a $100 Increase in Reserves

Bank A

Assets	Liabilities
$100	$100
($20 reserves	
$80 loans)	

Bank B

Assets	Liabilities
$80	$80
($16 reserves	
$64 loans)	

Bank C

Assets	Liabilities
$64	$64
($12.8 reserves	
$51.2 loans)	

Demand For Money

THE DEMAND FOR MONEY

- The main concern is:
  - How much of your financial assets you want to hold in the form of money, which does not earn interest, versus how much you want to hold in interest-bearing securities.
  - The demand for money is a demand for real balances.

KEYNESIAN THEORY OF DEMAND FOR MONEY

THE TRANSACTIONS MOTIVE

- There exists a *trade-off* between the liquidity of money and the interest income offered by other kinds of assets.

- The *transactions motive* is the main reason that people hold money—to buy things.

TRANSACTIONS DEMAND FOR MONEY

BAUMOL (1952)

Mr. A gets paid Rs. 1,800 per month. Mr. A spends his income evenly over the month (Rs. 60 per day).

Case I A holds Rs. 1,800 in cash and spend Rs. 60 per day.

Case II A keeps with him Rs. 60 for the first day and keeps the rest Rs. 1740 in bank as savings.

In this case then A will have to go to bank each and every day to withdraw Rs. 60 for that particular day.

Benefit: Mr. A earns interest on money retained with the bank each day by keeping less money with himself.

Cost: Time cost, transaction cost and inconvenience of the trips to the bank each and every day.

The more the number of trips to the bank, means the less individual A hold money with him and larger the amount of interest earnings.

$$\text{Real Money Balance}\left(\frac{M^d}{P}\right) = L(Y, r)$$

where $L_Y \rangle 0$ and $L_r \langle 0$

The higher the interest rate, more money to be parked with the banks and less money to be hold in hands for transactions purposes.

$$\text{Real Money Balance}\left(\frac{M^d}{P}\right) = L = k\,Y - \kappa\,r \quad k, \kappa \rangle 0$$

$'Y'$ being the money income and $'r'$ being the interest rate.

BOND PRICE AND INTEREST RATE

A bond of Rs. 1000 is issued for five years with a 5% interest rate, paid yearly. Then interest rate rises to 6%.

- If you want to sell this bond, who would buy it when it is paying 1% below market rates (5% vs. 6%)? You have to sweeten the deal so the buyer gets a market rate for the bond.

- You can't change the interest rate on the bond. That's fixed at 5%. You can, however, change the price you will take for the bond.

भारतीय स्टेट बैंक State Bank of India

VSP STEEL TOWNSHIP ,VIZAG SECTOR-1,
UKKUNAGARAM, VIZAG 530032
Tel: 2518796

सावधि जमा सूचना
(सावधि जमा रसीद के एवज में)
TERM DEPOSIT ADVICE
(In lieu of Term Deposit Receipt)

ANNAPURNA EDUCATIONAL AND CHARITABLE SOC
B C ROAD GAJUWAKA
OPP DAYAL NAGAR POLICE QTRS
VISAKHAPATNAM
Visakhapatnam

नामांकन /Nomination : पंजीकृत / Registered / अपंजीकृत / Not Registered 24/12/2020

दिनांक / Date :

प्रिय महोदय/महोदया, Dear Sir/Madam,

हमें यह पुष्टि करते हुए प्रसन्नता है कि आपकी निम्नलिखित राशि हमारे पास जमा है, भविष्य में, कृपया आपके पत्राचार में खाता क्रमांक का संदर्भ अवश्य दें, हमारे साथ बैंकिंग करने के लिए धन्यवाद। We have pleasure in confirming details of the following amount held in deposit with us. Please quote the account number in all correspondence. Thank you for Banking with us.

नाम / Name: ANNAPURNA EDUCATIONAL AND CHARITABLE SOCIETY & Regional Director, SRC, NCTE

सिफ संख्या/ CIF No.: 85175521122

पैन संख्या/ PAN NO. AAATA3096N

खाता संचालन की विधि:/ Mode of Operation : SINGLE

योजना/ Scheme : STD-PUB OTH UNI 181D-10YRS

खाता क्रमांक / A/c No.	सावधि /Term	ब्याज दर Interest @	मूल राशि Principal Amt.	जारी करने की तारीख Value Date	परिपक्वता की तारीख Maturity Date
39891014714	5 Y	5.4 %	INR 7,00,000.00	24.12.2020	24.12.2025

परिपक्वता राशि/ Maturity Value : INR 9,15,320.00

Annualised Yield (%): 6.15

Printed 1 Times

भवदीय / Yours faithfully,

R. Ravi

प्राधिकृत हस्ताक्षरकर्ता / Authorised Signatory

State Bank of India 20536 VSP Steel Township

कृपया पृष्ठ पलटें / P.T.O.

यह परक्राम्य लिखत नहीं है This is not a Negotiable Document

1. Name and address of the person

2. Amount of deposit in Figures and words

3. Date of deposit

4. Period of deposit

5. Interest Rate

12. Terms of Deposit

THE KARNATAKA BANK LTD.
Regd. & Head Office: MANGALORE - 575 002
ದಿ ಕರ್ನಾಟಕ ಬ್ಯಾಂಕ್ ಲಿ. दि कर्णाटक बैंक लि.

KBL/MPL/TDR/C 442339

BRANCH: B LORE JAYANGR IX BLK (10)

NOT TRANSFERABLE

ABHYUDHAYA CASH CERTIFICATE → (8) 061098796

Received from (1)
1.MR SOMAYAJI K V R
NO97 .3RD MAIN .2ND CROSS .SARAKKI DOLLERS
LAYOUT .J P NAGAR .4TH PHASE .BANGALORE-78
BANGALORE - 560078

the sum of Rs. (2) Rs.1.000.00 (One Thousand only.)

L.F. No.	Nature of Deposit and Receipt No.	(3) Issue Date	(4) Period	(5) Interest Rate	(6) Maturity Date	Nomination Registration No.
	0611500218414601 W.E.F 29-04-2009	29-04-2009	Days : Months: Years : 1 Yrs	8.50	29-04-2010	

Repayable to: SELF (7)

If Minor, Date of Birth

Maturity Value Rs.: Rs.1.088.00 (One Thousand Eighty Eight only.) (9)

(12) THIS IS REINVESTMENT DEPOSIT WITH COMPOUNDING OF INTEREST QUARTERLY

1. MATURITY VALUE MENTIONED HEREIN MAY VARY DUE TO DEDUCTION OF TAX AT SOURCE AT THE APPROPRIATE RATE AS PER INCOME TAX RULES UNLESS OTHERWISE ELIGIBLE FOR EXEMPTION AND COMPOUNDING EFFECT OF INTEREST ON THE AMOUNT OF TAX SO DEDUCTED.
2. PENALTY LEVIED FOR PREMATURE CLOSURE.

(11)

Authorised Signatory — ...AM PRASAD, Manager P.A. No.765

Authorised Signatory — R.S. Pandurangaa P.A. No: 2391

F. No. 691/9703 Interest ceases from maturity date. Please present this duly discharged for renewal/payment. *Preferential rate is applicable only if the deposit is continued for the full contract period and in case of premature closure, interest shall be paid as per the applicable...

10. Name of branch

8. Type of deposit

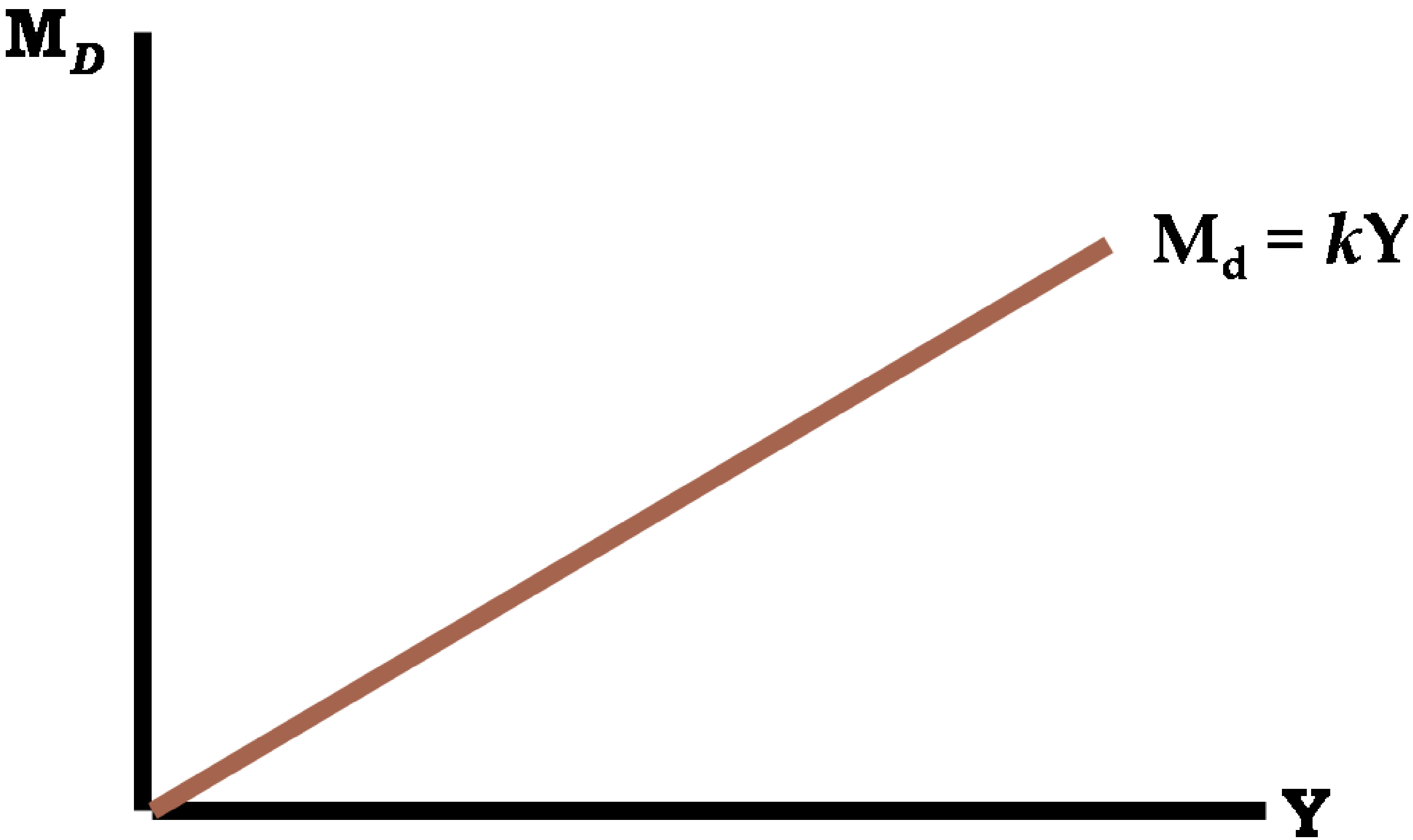
6. Maturity (Due) Date

7. Payable to whom

9. Maturity value

11. Signature of Manager

Money Demand and Income



WALRAS' LAW

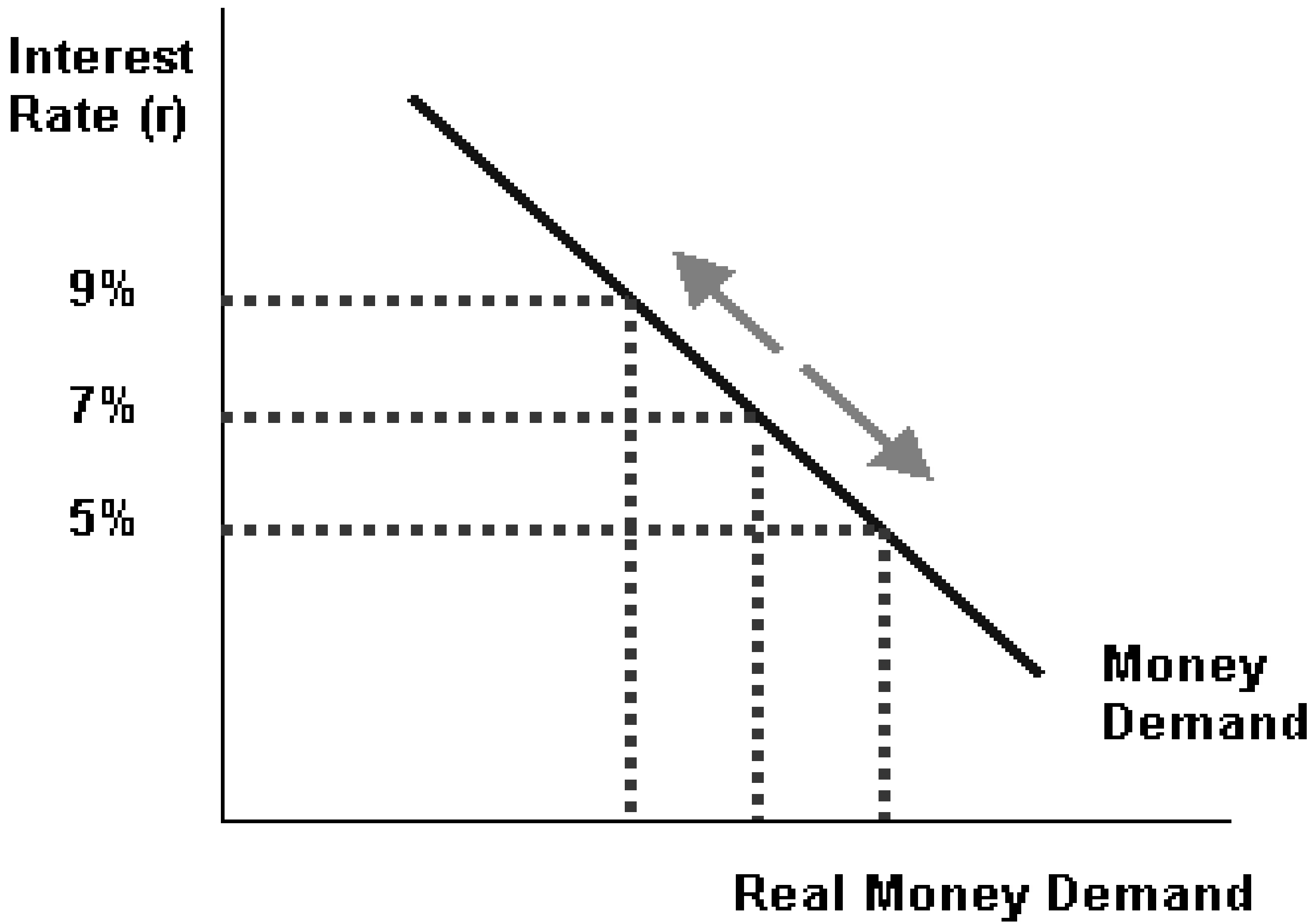
In an economy of n markets, when (n–1) markets clear, the n^{th} market will also be in equilibrium.

Total supply of money and bonds [M+B] must be equal to the total demand for money and bonds ($M^D + B^D$), since people who are trying to sell bonds are trying to obtain money.

$$M + B = M^D + B^D$$

$$\text{or} \quad (M^D - M) + (B^D - B) = 0$$

If there is an excess demand for money, there must be an excess supply of bonds.


Interest Rate (r)
9%
7%
5%
Money Demand
Real Money Demand

THE PRECAUTIONARY MOTIVE

- Cash balances held in case of unforeseen outlays, essentially of a transaction nature (e.g. ***unforeseen medical bill***).

- People hold money because they can't anticipate every need, there is uncertainty, so they hold more.

SPECULATIVE DEMAND FOR MONEY

- Speculation = buying an asset in the hopes that its price will rise, e.g., a bond.
- Bond price vary inversely with interest rate, i.e., *if interest rate rises, bond price falls*.
- So, lower the interest rate, the more you might expect them to rise and bond prices to fall, so you would hold fewer bonds and more money.